# lowThrust — how it works

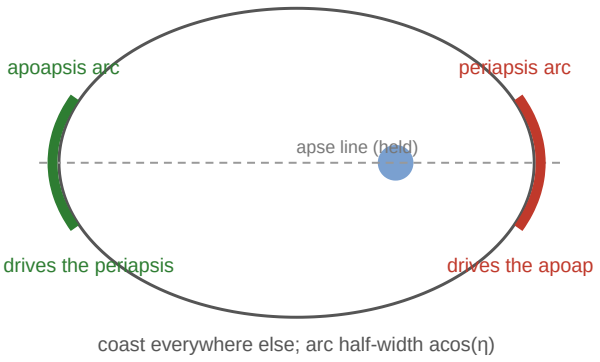
Low-thrust transfers within one SOI, to a moon, to another planet or a planet's moon: the solver, its phase machine and the stand-in solar system it flies in. No kRPC side yet.

## What it does

Given the vessel's current orbit and a target orbit (apoapsis, periapsis, inclination, longitude of the ascending node, about any body; anything left out is free), it decides every tick whether to thrust, which way and how hard, until the target orbit is reached. Every decision is made from the current state and a fresh prediction; there is no stored schedule, so a weak or lost tick is made up by the same law later.

| module | role |
|---|---|
| `orbit.py` | elements of any conic, exact propagation (universal variables), time to periapsis, Lambert, radius toward a fixed direction |
| `guidance.py` | the within-SOI law: `lock` (apse line and target plane), `command`, `arrived` |
| `plan.py` | flies the law in the simulator to pick the arc width that meets a deadline |
| `sim.py` | single-body stand-in; the vessel (constant mass, thrust, Isp); shared RK4 step and tick |
| `bodies.py` | stock bodies on Keplerian rails with their spheres of influence |
| `transfer.py` | targeting (closest approach, B-plane miss, correction, moon push, window, parking orientation, Lambert re-target, hazards) and the capture law |
| `mission.py` | the phase machine of a whole transfer in patched conics; routing and leg chaining |
| `tools/` | `simfly.py` (one SOI, deadline), `simmission.py` (a transfer: timeline, burns, dv), `sweep.sh` (destination × thrust × date grid) |

## The within-SOI law: Hohmann in many small burns




Two arcs centred on a fixed apse line; each fixes the opposite apsis.

**Decoupled.** A tangential burn at periapsis moves the apoapsis and, to first order, not the periapsis, and the other way round. The quantity driven is the radius toward the far end of the apse line, r = p / (1 + e·u), smooth through circular; its rate per m/s of prograde is differenced numerically (`RATE_DV`).

**Arcs** are ±acos(η) wide, η the off-apsis effectivity (default 0.99, ±8°). A burn starts half its remaining duration before the apsis, runs at full throttle until the radius is on target or the arc ends; the last tick's throttle is the fraction still needed. Prograde or retrograde only, so plain SAS modes can hold it.

**Order:** outward first (raise the apoapsis before the periapsis), inward first when lowering; the apoapsis is not lowered while the plane still has to change.

**Apse line:** held through each burn, re-locked to the osculating periapsis while coasting; a target with a held line (the parking ellipse) keeps it fixed and corrects it at quadrature once the apsides are close.

### Plane change

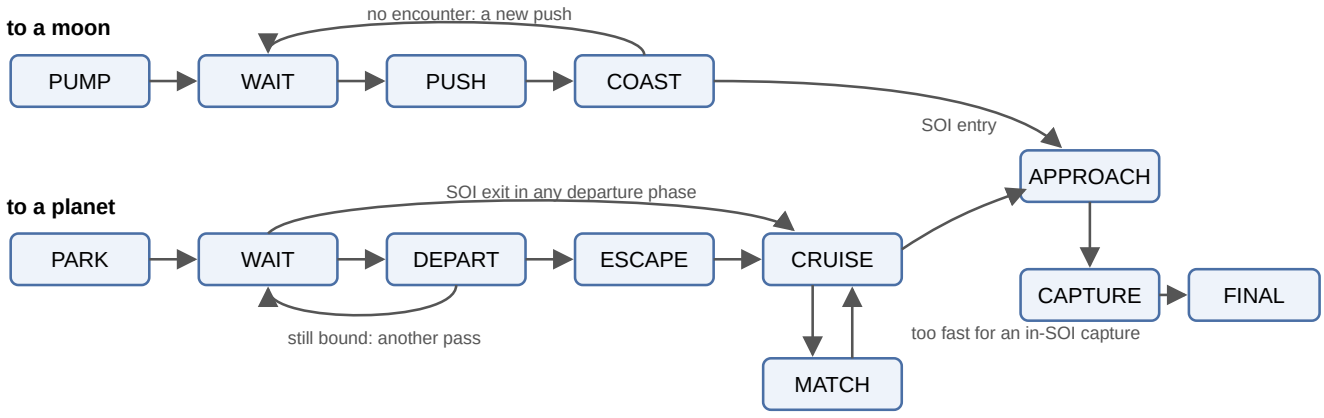
The target plane is fixed once (`lock`). Normal/anti-normal arcs of the same width at the two nodes with it, burn time $\delta \cdot h/(r_{node} \cdot a)$; a node whose radius is below η of the other's is skipped (a plane change costs the speed it turns, least where the vessel is slowest).

| gate | effect |
|---|---|
| apoapsis still being raised | no plane burns yet: the plane is changed on the biggest orbit |
| plane still to change | the apoapsis is not lowered |
| periapsis still to change (measured as the apsis arc measures it — toward a held apse line when there is one) | plane burns only at the node on the apoapsis side (within an arc of quadrature allowed), where they share the periapsis burn |
| apsis arc and node arc both due | one burn: direction (prograde, normal) in proportion to the seconds each still needs, started half the combined duration early |

### The planner and the time budget

`plan.plan` flies the law in the simulator at `PLAN_GRID` (13) arc widths between `ETA_MAX` (0.999, ±2.6°) and `ETA_MIN` (0.2, ±78°) and takes the narrowest — cheapest — that arrives within the budget; without one it flies `ETA_DEFAULT`. The prediction is the law itself flown forward.

## Transfers between bodies: the phase machine

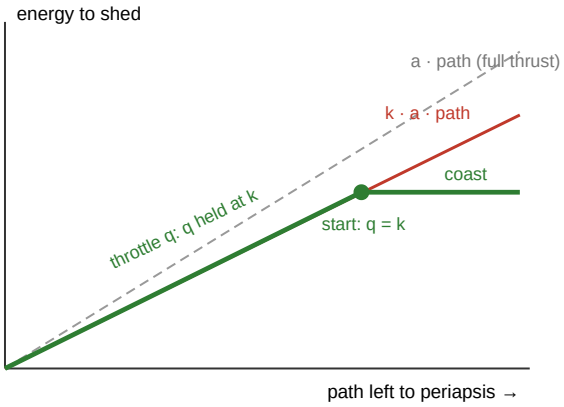



TRIM (a phasing burn at periapsis) and an evasive burn in an unplanned SOI can interrupt any phase. A planet's moon is two legs.

The phases of a transfer (`mission.Mission`), each re-decided every tick from the current state.

| phase | what it does | ends when |
|---|---|---|
| PUMP | within-SOI law to the waiting orbit: in the moon's plane, apoapsis 0.6 of the way, periapsis as low as possible but high enough for the engine to capture in one pass and outside other moons' bands. With another moon inside the target's, the whole waiting orbit stays below that moon's band, periapsis raised until the push across it fits one pass. | reached |
| WAIT (moon) | once a revolution, ahead of periapsis: solve the prograde push there that puts a direct encounter (within 0.75 revolution) on the aim point. Skip the pass if its coast meets another moon, or if it arrives faster than the capture takes (the bar rises 25% per skip). | a push exists, periapsis half its duration away |
| PUSH | prograde on a ledger (planned dv less delivered), refined by local re-solves | ledger spent |
| COAST / CRUISE | B-plane corrections when one needs more than 0.5 m/s, rechecked every tenth of the time left. To a moon, no encounter goes back to WAIT. To a planet, a miss beyond min(20 SOI, 1% of the target's distance) or a correction over 150 m/s is re-planned as a Lambert arc; the encounter is looked for round the planned arrival. Triggers MATCH when planned. | SOI entry |
| PARK | within-SOI law, held apse line, to the parking ellipse oriented on the window's departure asymptote | reached |
| WAIT (planet) | coast to the periapsis pass nearest the window; skip a pass whose departure burn, flown forward on RK4 as DEPART will fly it, meets a moon | half the push before periapsis |
| DEPART | prograde until the hyperbolic excess is the window's, or 60° past periapsis (signed anomaly) | escaping: ESCAPE; still bound: WAIT |
| ESCAPE | coast out; a bound orbit whose periapsis fell below the floor is re-parked | SOI exit → CRUISE |
| MATCH | brakes about the star onto an aim point just outside the destination's SOI (below) | 600 s before the aim point → CRUISE |
| APPROACH | nudge the hyperbola's periapsis onto the capture altitude along its gradient; coast an unplanned flyby | the capture law says burn |
| CAPTURE | the capture law (below); coasts through another body's SOI and re-decides on exit | bound (FINAL) / lost (fails) |
| FINAL | within-SOI law to the target orbit | arrived |

## The capture law




q = energy to shed / (a × path left to periapsis). Coast until q reaches k, then throttle at q: q stays constant and the last of the energy goes at periapsis.

The capture only has to make the orbit **bound**: below 0.3 of the SOI (or the final apoapsis, if higher), or, once past periapsis, anything below the highest apoapsis clear of the planet's moons — and always with the periapsis above the safe floor (30 km over the atmosphere). FINAL then shapes the orbit.

**Short:** if what is left fits within 80% of the time spent within ±60° of periapsis, the burn is centred on periapsis at full throttle.

**Long:** thrust along the velocity removes specific energy at exactly *a* per metre of path flown, wherever it is flown. So the law coasts until the energy above a bound orbit reaches `CAPTURE_PATH_SHARE` (k = 0.8) of what full thrust can remove on the path left to periapsis, then throttles at that ratio — a steady throttle near k, the rest in reserve. Outbound and still unbound: full throttle.

**Direction:** retrograde (sized to one tick's dv), plus a periapsis-gradient term that holds the periapsis between the floor and 1.5× the aimed one (`CAPTURE_RP_BAND`); below the floor, only the lift. Steps change the velocity by at most 5%.

It runs only inside the destination's SOI, where the destination alone pulls. Planning flies the same law (`transfer.fly_capture`) from the window's arrival to choose between a capture and MATCH.

## MATCH: braking about the star

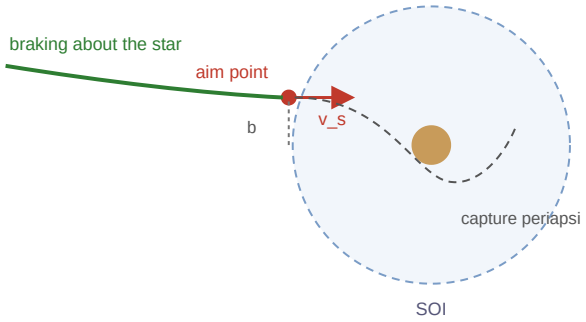
**When:** planned when the arrival is faster than a capture starting within 0.8 of the SOI can take. Its arrival speed $v_s$ is 0.8 (`ARRIVE_MARGIN`) of the fastest the capture takes — a long capture inside the SOI, or one pass high and loose, whichever is faster.

**Start:** from the cruise's predicted encounter (arrival speed v, time to closest approach): when braking from v to $v_s$ at `MATCH_SHARE` (0.8) of the engine, plus the standoff, needs all the time left.

**Law:** zero-effort-miss / zero-effort-velocity guidance, $a = 6\,ZEM/t^2 - 2\,ZEV/t$. ZEM and ZEV come from coasting the vessel and the destination t ahead about the star. t is the time a constant deceleration from v takes to reach the aim point, $t = 2\,(v \cdot t_{ca} - d_{back}) / (v + v_s)$, recomputed every tick from a fresh encounter.

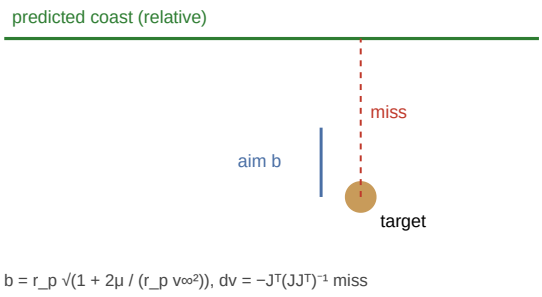
**Aim point:** `MATCH_STANDOFF` (1.05) SOI radii out, moving at $v_s$ along the arrival direction, offset by the impact parameter of the capture periapsis on the B-plane side the cruise aims for.

**End:** 600 s before the aim point (`MATCH_END_S`), or once $v \le 1.05\,v_s$; CRUISE then carries it into the SOI.




MATCH ends at the aim point at $v_s$ with the capture's impact parameter b; the capture law takes over inside.

## Targeting




The B-plane miss and the minimum-norm correction.

**Closest approach** of the exact two-body coast to the target on its rails: a coarse scan then golden section, or a window round a given UT (the planned arrival). A minimum at the window's edge is no encounter.

**Aim point:** the impact parameter b for the capture periapsis, on the side whose passage is prograde about the target's orbit normal. **Correction:** numerical Jacobian of the two miss components against the three impulse axes, minimum-norm.

**Moon push:** the smallest prograde impulse at periapsis whose signed closest-approach distance equals b. **Re-target:** Lambert to the aim point over arrival times 0.6–1.6 of the planned time left, cheapest in impulse plus arrival speed, held while burning.

## Departure, capture design and hazards

**Window:** Lambert over departure dates (one synodic period from the pump's estimated end) and flight times (0.5–1.5 of the Hohmann's), least |v∞ out| + |v∞ in|. **Parking ellipse:** periapsis above the atmosphere, apoapsis 0.9 of the highest clear of the planet's moons; its plane contains the departure asymptote, the one cheapest to reach. A weak engine's departure takes several periapsis passes; whatever v∞ it leaves with, the cruise's re-target builds the rest about the star.

**Capture design:** the lowest capture periapsis (≥ the target's, ≥ the floor) at which one pass sheds the arrival speed into the preferred ellipse (0.3 SOI, under the moons) or a looser one; failing that, a long capture inside the SOI if the planning prediction says it starts within 0.8 SOI; failing that, MATCH. Refused only if even an arrival at rest cannot be captured.

**Hazards:** each arc burn's pass dv is capped at the largest fraction whose next revolution stays out of every other moon's SOI; a phasing trim of ±0.5–8 m/s at periapsis if the coming revolution would meet a moon; an evasive burn along the periapsis gradient inside an unplanned SOI with the periapsis below the floor; after a slingshot, braking until the apoapsis is inside 0.6 SOI.

## The stand-in

Patched conics: the vessel's state is relative to the body whose SOI it is in. A coasting tick is propagated exactly (universal variables, starting from whichever of the short- and long-arc guesses is closer) and bisected to an SOI boundary when it crosses one; the frame then switches by the body's state on its rails. Thrusting ticks are `THRUST_SUBSTEPS` (4) RK4 steps. Ticks are uniform in angle ($r^2/h$ per 2°, capped at period/180). The vessel has constant mass and unlimited propellant — the stand-in plans paths; dv is thrust/mass × burn time and xenon is reported from it with the rocket equation. A flight ends on arrival, impact, the time limit (`MISSION_MAX_DAYS`) or a failed capture.